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SUPERSEDING
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MILITARY SPECIFICATION

# CATAPULT, AIRCRAFT EJECTION SEAT, M3A1 ASSEMBLY

## 1. SCOPE

1.1 This is the applicable detail specification for Catapult, Aircraft Ejection Seat, M3A1, Assembly, loaded with Cartridge, Aircraft Ejection Seat, Catapult, M36, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

### SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons, General Specification for

MIL–I–45208 —Inspection Requirements, General Specification for

### STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–109 —Quality Assurance Terms and Definitions

MIL–STD–414 —Sampling Procedures and Tables for Inspection by Variables for Percent Defective

### DRAWINGS

MUNITIONS COMMAND

MS–28741 —Hose Assembly—Detachable End Fitting, Medium Pressure

F8593635 —Catapult, Aircraft Ejection Seat, M3-A1, Assembly

F8797431 —Ammunition Container, Fiber M304 for Catapult, Aircraft Ejection Seat, M3A1

F8797432 —Box, Packing Ammunition for Catapult, Aircraft Ejection Seat, M3A1

IEL–8593635 —Index to Inspection Equipment List for Catapult, Aircraft Ejection Seat, M3-A1, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the

FSC 1375



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extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

INTERSTATE COMMERCE COMMISSION—49 CFR 71–78. The Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles. (The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71–78".)

## 3. REQUIREMENTS

**3.1 General.** Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

**3.2 Torque.** Torque requirements shall be as specified in table I. No loosening or deformation shall occur upon application of the breakaway torque.

TABLE I.

| *Component* | *Seating or bottoming (inch pounds)* | *Breakaway (inchpounds), min.* |
|---|---|---|
| Block and Inside Tube after Insertion of Cartridge | 400 + 50 | 300 |
| Stop Ring and Trunion | 550 + 50 | 400 |

**3.3 Assembly**

3.3.1 The flange face surface of the plug shall be in complete contact with the cartridge.

3.3.2 The block shall be free to rotate 360 degrees.

**3.4 Residual magnetism.** The catapult shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

**3.5 Functional.**

3.5.1 *Ballistics.* Catapults loaded with accepted M36 cartridges shall propel a 350 pound mass vertically and shall comply with the requirements listed in Table II.

3.5.2 The catapult shall not fail to fire.

3.5.3 The catapult shall function without mechanical failure (see 6.1).

**3.6 Workmanship.** The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 *Burr.* No part shall have a bur which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

TABLE II.

| Temperature | Velocity min. | Acceleration max. | Rate of change of acceleration, max. | Ignition delay max. (see 6.2) |
|---|---|---|---|---|
| —65° to —70° F. | 70 fps | 24g | 175 g/sec. | 150 milliseconds |
| 65° to 75° F. | 77 fps | 20.5g | 180 g/sec. | 150 milliseconds |
| 160° to 165° F. | 82 fps | 24g | 250 g/sec. | 150 milliseconds |

3.6.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the government. Inspection records of the examination and tests shall be



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kept complete and available to the government as specified in the contract or order. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be made in accordance with MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 **Inspection provisions.**

4.2.1 *Lot.*

4.2.1.1 *Presentation of lots.* The product shall be presented in accordance with MIL–STD–105 or MIL–STD–414, as applicable.

4.2.1.2 *Size of lot.* A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 *Examination.* One hundred percent examination shall be performed for all critical critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in table III. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of table III. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III.

| Lot size | Sample size | Major | | Minor | |
|---|---|---|---|---|---|
| | | Accept | Reject | Accept | Reject |
| 1—300 | 100% Examination | | | | |
| 301—800 | 50 | 0 | 1 | 1 | 2 |
| 801—1800 | 90 | 1 | 2 | 2 | 3 |

4.2.2.1 *Cartridge, Aircraft Ejection Seat, Catapult, M36, Dwg. 8593632, covering a detail of Dwg. F8593635.*

| | *Defect* | *Method of inspection* |
|---|---|---|
| *Critical* | | |
| 1. | Mixed cartridges | Visual |
| 2. | Metal defective | Visual |
| 3. | "O" ring missing | Visual |
| 4. | Sealing disc missing, damaged or improperly secured | Visual |
| 5. | Marking missing, incorrect or illegible | Visual |
| *Major* | | |
| 101. | "O" ring split, cut or cracked | Visual |
| 102. | Foreign matter | Visual |
| *Minor* | | |
| 201. | Case scratched, wrinkled, bulged or dented | Visual |
| 202. | Varnish missing, or inadequate | Visual |

4.2.2.2 *Seal, Ordnance, Dwg. C8593665, covering a detail of Dwg. F8593635.*



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| Defect | Method of inspection |
|---|---|
| *Critical:* None defined. | |
| *Major:* None defined. | |
| *Minor* | |
| 201. Inside diameter under flange | Gage |
| 202. Inside diameter adjacent to flange | Gage |
| 203. Radius missing or incorrect | Visual |
| 204. Metal defective | Visual |
| 205. Protective finish, inadequate or defective | Visual |
| 206. Bur | Visual |
| 207. Foreign matter | Visual |

4.2.2.3 *Assembly, Dwg. F8593635, less all D wgs. except C8593643, C8593644, F8593648, B-8593654, C8593656, C8593657 and AN6227–11.*

| Defect | Method of inspection |
|---|---|
| *Critical* | |
| 1. Gas port hole obstructed | Visual |
| 2. Any part missing | Visual |
| 3. Improper assembly | Visual |
| *Major* | |
| 101. Thread damaged | Visual |
| 102. Bur | Visual |
| 103. Foreign matter | Visual |
| *Minor:* None defined. | |

4.2.2.4 *Assembly, Dwg. F8593635, less Dwgs. C8593639, D8593640, D8593649, C8593650, C8593651 and B8593660.*

| Defect | Method of inspection |
|---|---|
| *Critical* | |
| 1. Flange face surface of plug not in complete contact with cartridge | Gage |
| 2. Cartridge missing | Visual |
| 3. Seals on inside tube missing | Visual |
| *Major* | |
| 101. Shock washer missing | Visual |
| 102. Bur | Visual |
| 103. Foreign matter | Visual |
| *Minor:* None defined. | |

4.2.2.5 *Assembly, Dwg. F8593635, less all Dwgs. except C8593639, D8593640, D8593649, C8593650, C8593651 and B8593660.*

| Defect | Method of inspection |
|---|---|
| *Critical* | |
| 1. Seal on trunion missing | Manual/Visual |
| *Major* | |
| 101. Bur | Visual |
| 102. Foreign matter | Visual |
| *Minor:* None defined. | |

4.2.2.6 *Assembly, Dwg. F8593635 (with stop ring not engaged with trunnion).*



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| Defect | Method of inspection |
|---|---|
| *Critical* | |
| 1. Tubes not free to slide | Manual |
| *Major:* None defined. | |
| *Minor:* None defined. | |

4.2.2.7 *Assembly, Dwg. F8593635.*

| Defect | Method of inspection |
|---|---|
| *Critical* | |
| 1. Mixed catapults | Visual |
| 2. Marking missing, incorrect or illegible | Visual |
| *Major* | |
| 101. Total length, max. | Gage |
| 102. Diameter of mounting hole | Gage |
| 103. Length from cap to mounting hole | Gage |
| 104. Length between mounting points | Gage |
| 105. Block not free to rotate 360 degrees | Manual |
| 106. Bur | Visual |
| 107. Foreign matter | Visual |
| *Minor* | |
| 201. Location of labels | Visual |
| 202. Shipping plug missing | Visual |
| 203. Ordnance seal missing or insecure | Visual |
| 204. Protective finish, inadequate or defective | Visual |

4.2.2.8 *Fiber container, Dwg. F8797431.*

| Defect | Method of inspection |
|---|---|
| *Critical:* None defined. | |
| *Major:* None defined. | |
| *Minor* | |
| 201. Gap between body and cover | Manual |
| 202. Contents loose | Manual |
| 203. Metal ends loose or distorted | Manual |
| 204. Cut or damaged through all imprenated layers | Visual |
| 205. Marking missing, incorrect or illegible | Visual |
| 206. Tape incomplete or badly wrinkled | Visual |
| 207. Length of tear tab, min. | Visual |
| 208. Color of tape, incorrect | Visual |

4.2.2.9 *Box, packing, Dwg. F8797432.*

| Defect | Method of inspection |
|---|---|
| *Critical:* None defined. | |
| *Major:* None defined. | |
| *Minor* | |
| 201. Contents loose | Manual |
| 202. Hardware missing, loose, broken or improperly engaged | Manual/Visual |
| 203. Strapping missing, loose, broken or improperly assembled | Manual/Visual |
| 204. Contents of box exposed | Visual |
| 205. Box damaged | Visual |
| 206. Marking missing, incorrect or illegible | Visual |
| 207. Number of units, incorrect | Visual |
| 208. Car seal missing, unsealed or improperly positioned | Visual |



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4.2.3 *Inspection testing.* The following tests shall be performed in accordance with 4.3, except as provided in MIL–I–45208.

4.2.3.1 *Breakaway torque.* A breakaway torque shall be applied to each catapult threaded joint listed in table I. Failure of any catapult to comply with the requirements of 3.2 shall be cause for rejection of the catapult.

4.2.3.2 *Residual magnetism.* Each assembly shall be tested for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.4 shall be cause for rejection of the catapult.

4.2.3.3 *Functional.* The following tests shall be performed by the Government.

4.2.3.3.1 *Ballistics.* Twenty-one units shall be tested to determine compliance with the requirements of 3.5.1, 3.5.2 and 3.5.3. The units shall be temperature conditioned as provided in table IV and 4.3.3.1.1. The variables sampling plan of MIL–STD–414, example C–2, for the AQL listed in table IV shall be used for each sample of 7 units. Failure of any unit to comply with the requirements of 3.5.1, 3.5.2 and 3.5.3 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

TABLE IV.

| Temperature | Sample size | AQL |
|---|---|---|
| —65° to —70° F. | 7 | 0.65% |
| 65° to 75° F. | 7 | 0.65% |
| 160° to 165° F. | 7 | 0.65% |

4.2.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8593635, except as provided in MIL–I–45208.

**4.3 Test methods and procedures.**

4.3.1 *Breakaway torque.* A torque wrench set at the torque requirement specified in 3.2 shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to application of the torque.

4.3.2 *Residual magnetism.* The assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the catapult 6 inches from and at the same height as the compass in a north-south horizontal position (trunnion end at south) with the compass set in an east-west heading as shown in figure 1. The assembly shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

4.3.3 *Functional.*

4.3.3.1 *Ballistics.* The catapult shall be rigidly supported in the fixture by the mounting points and fired with an M3A1 Initiator connected to the catapult with 10 feet of aircraft hose, size 4, as specified in MS–28741. The catapult shall comply with the requirements of 3.5. The catapult shall be tested within 10 minutes after removal from the conditioning chamber. When this time limit is exceeded, the catapult shall be reconditioned at the original temperature.

4.3.3.1.1 *Temperature conditioning.* A standard M3A1 Catapult, loaded with a dummy M36 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

## 5. PREPARATION FOR DELIVERY

**5.1 Packing and marking.** Packing and marking shall be in accordance with Drawings F8797431 and F8797432.

**5.2 Delivery.** Delivery shall be in accordance with Interstate Commerce Commission Regulations cited in 2.2.

**5.3 Data cards.** Data card information shall be as specified in Specification MIL–A–2550.



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## 6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is definied as the time that elapses between the firing of the cartridge primer and the beginning of the continuous rise on the thrust time record.

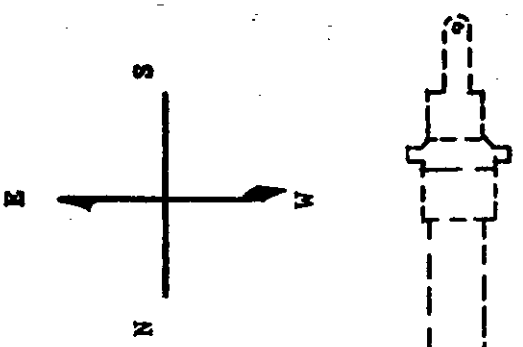
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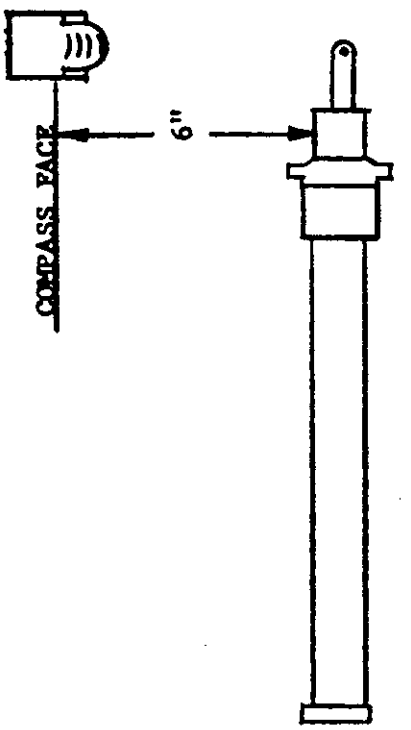


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TO THIS POSITION

PLAN VIEW




MOVE FROM THIS POSITION

NOTE: CATAPULT AND COMPASS SHALL LIE IN A HORIZONTAL PLANT

LAYOUR FOR DETERMINATION OF RESIDUAL MAGNETISM IN CATAPULT, AIRCRAFT EJECTION SEAT, M3A1

FIGURE 1. *Pilot's standby compass, type AN–5766.*



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